

Astronomy Question Bank

These questions were carefully written or selected to prepare students for the astronomy team selection process in Brazil. Some questions are not original; those are marked with () before the statement. The question-bank template is the same one used by Professor Kevin Zhou. His work is invaluable, and many ideas in this set can be found in his handouts.

1 Photometry and Modern Physics

- [3] **Problem 1.** The goal of this question is to derive an expression for the *optical depth*. Imagine that a beam of light passes through a region of space containing a certain number of particles. If I_0 is the intensity of the light before it passes through that region of space and I is the intensity afterward, the optical depth is defined as:

$$I = I_0 e^{-\tau}$$

where τ is the optical depth.

Consider a region of space with particle number density n .

- What is the number of particles in an area A and thickness dz ?
- Assuming that each particle has cross section σ , what is the area blocked by the particles?
- Find the formula for τ .

Solution

- a) The volume number density of particles is n , so the number dN of particles in a volume dV is

$$dN = n dV \equiv n A dz$$

- b) If each particle occupies an area σ , the area occupied by dN particles is

$$dS = \sigma dN = \sigma n A dz$$

- c) The area occupied by the particles is expected to obstruct the passage of light. Since the intensity is proportional to the area available for the light to pass through, we have

$$\frac{dI}{I} = -\frac{dS}{A} = -n\sigma dz$$

Integrating, we obtain

$$I = I_0 e^{-n\sigma z}$$

That is,

$$\tau = n\sigma z$$

- [2] **Problem 2.** The Triangulum Galaxy, $M33$, is the third-largest galaxy in the Local Group. It lies at a distance $d = 970$ kpc from us and has an apparent magnitude of 5.72. Given that it contains approximately 40 billion stars, find the average luminosity of the stars in $M33$. Does your estimate seem consistent with reality? Why?

Solution

Using Pogson's equation

$$m - M_{\odot} = -2.5 \log \left(\frac{L_g}{L_{\odot}} \left(\frac{10 \text{ pc}}{d} \right)^2 \right)$$

Solving for L_g ,

$$L_g = L_{\odot} \left(\frac{d}{10 \text{ pc}} \right)^2 10^{-0.4(m - M_{\odot})}$$

We also have $L_g = \overline{L}_{\star} N$, and thus

$$\overline{L}_{\star} = \frac{L_{\odot}}{N} \left(\frac{d}{10 \text{ pc}} \right)^2 10^{-0.4(m - M_{\odot})}$$

Using the values from the table of constants, we obtain

$$\overline{L}_{\star} \approx 0.1 L_{\odot}$$

This estimate does not match reality, since factors such as interstellar extinction increase the apparent magnitude of $M33$. A consistent estimate would be of the same order of magnitude as the luminosity of the Sun.

- [4] **Problem 3.** Consider a universe with a constant stellar number density, that is, a number of stars per unit volume, of value n . Assume that every star has $L = L_{\odot}$ and that the universe contains a total of N_0 stars.

- What is the probability that the absolute magnitude of a star, seen from the center of the universe, lies between m and $m + dm$, where dm is an infinitesimal interval of magnitude? Leave your answer in terms of m and the largest possible magnitude, m_{lim} , of a star at the "edge" of the universe.
- What is the probability that a star can be observed with the naked eye?

Data:

$$\int_{-\infty}^6 10^{0.6(x-A)} dx \approx 2881.6 e^{-1.381A}$$

Solution

- The apparent magnitude of a star is given by

$$m = -2.5 \log F + C$$

where C is a constant. Writing $F = L/4\pi r^2$, we have

$$m = -2.5 \log \left(\frac{L}{4\pi r^2} \right) + C = 5 \log r - 2.5 \log \left(\frac{L}{4\pi} \right) + C$$

Since L is the same for every star, we can absorb the second term into the constant. Thus

$$m = 5 \log r + C \rightarrow r = 10^{0.2(m-C)}$$

Differentiating the expression for m , we have

$$\frac{dm}{dr} = \frac{5}{r \ln 10}$$

Now we find the probability that a star lies at a distance r .

The number of stars between r and $r + dr$ is

$$dN(r) = 4\pi n r^2 dr$$

Dividing by N_0 , the probability that a star lies at that distance is

$$dP(r) = \frac{dN(r)}{N_0} = \frac{4\pi n r^2}{N_0} dr \rightarrow dr = \frac{N_0}{4\pi n r^2} dP(r)$$

Substituting the expression for dr ,

$$\frac{r \ln 10}{5} dm = \frac{N_0}{4\pi n r^2} dP(r)$$

Isolating $dP(r)$,

$$dP(r) = \frac{4\pi \ln 10 n r^3}{5 N_0} dm$$

Substituting the expression for r , we change variables $r \rightarrow m$ in dP :

$$dP(m) = \frac{4\pi n \ln 10}{5 N_0} 10^{0.6(m-C)} dm$$

To find C , compare with the limiting magnitude at the "edge" of the universe. Since the universe contains N_0 stars, its radius is given by

$$N_0 = \frac{4\pi R^3}{3} \rightarrow R = \left(\frac{3N_0}{4\pi} \right)^{1/3}$$

Writing the equation for m_{lim} ,

$$m_{lim} = 5 \log R + C$$

we can isolate the constant $C = m_{lim} - 5 \log R$. Substituting into the expression for $dP(m)$,

$$dP(m) = \frac{4\pi n \ln 10}{5N_0} 10^{0.6(m-m_{lim}+5 \log R)} dm$$

Working through this expression,

$$dP(m) = \frac{4\pi n \ln 10}{5N_0} 10^{0.6(m-m_{lim})} 10^{3 \log R} dm$$

Using the logarithm identities $a \log b = \log b^a$ and $10^{\log x} = x$, we have

$$dP(m) = \frac{4\pi n R^3 \ln 10}{5N_0} 10^{0.6(m-m_{lim})} dm$$

Substituting R ,

$$dP(m) = \frac{3 \ln 10}{5} 10^{0.6(m-m_{lim})} dm$$

The probability therefore depends on neither n nor N_0 .

b) For a star to be visible, its magnitude must satisfy $m \leq 6$. Thus

$$P(m \leq 6) = \int_{-\infty}^6 dP(m) = \frac{3 \ln 10}{5} \int_{-\infty}^6 10^{0.6(m-m_{lim})} dm$$

Using the integral given in the statement,

$$P(m \leq 6) = \frac{3 \ln 10}{5} 2881.6 e^{-1.381 m_{lim}}$$

Simplifying the numerical factors,

$$P(m \leq 6) \approx 3981.08 e^{-1.381 m_{lim}}$$

Out of curiosity, taking $m_{lim} \approx 40$ as an estimate, we would have

$$P(m \leq 6) \approx 10^{-20}$$

Even in the simplest model of the universe, our capability and our insignificance prevail.

[4] **Problem 4.** The *surface brightness*, flux per solid angle per frequency, B_ν , is given by Planck's

law:

$$B_\nu = \frac{dF}{d\nu d\Omega} = \frac{2h\nu^3}{c^2(e^{\frac{h\nu}{k_B T}} - 1)}$$

a) Find an expression for:

$$B_\lambda = \frac{dF}{d\lambda d\Omega}$$

b) Starting from B_λ , find the peak wavelength λ_{max} emitted by a star of temperature T (you will have to solve something numerically).

c) At low frequencies we have the Rayleigh-Jeans approximation. Obtain an expression for B_ν at low frequencies.

Solution

a) We have

$$B_\lambda = \frac{dF}{d\lambda d\Omega} = \frac{dF}{d\nu d\Omega} \frac{d\nu}{d\lambda}$$

Using the fundamental wave relation $c = \nu\lambda \rightarrow \nu = c/\lambda$, we arrive at

$$B_\lambda = \frac{2h(c/\lambda)^3}{c^2(e^{\frac{hc}{\lambda k_B T}} - 1)} \frac{d}{d\lambda} \left(\frac{c}{\lambda} \right) = \frac{2hc^2}{\lambda^5} \frac{1}{(e^{\frac{hc}{\lambda k_B T}} - 1)}$$

b) To solve this part we need the maximum of B_λ . Differentiating B_λ directly is extremely tedious. A more efficient approach is to take the natural logarithm of B_λ and differentiate that. At a maximum, both procedures give the same result.

$$\ln B_\lambda = -5 \ln \lambda - \ln(e^{hc/\lambda k_B T} - 1) + C$$

where C is a constant that absorbs the $\ln$ of the other constants in B_λ . Differentiating further,

$$\frac{d \ln B_\lambda}{d\lambda} = -\frac{5}{\lambda} + \frac{e^{hc/\lambda k_B T}}{e^{hc/\lambda k_B T} - 1} \frac{hc}{k_B T \lambda^2} = 0$$

Defining $x = \frac{hc}{k_B T \lambda}$,

$$\frac{x e^x}{e^x - 1} = 5$$

$$x = 5e^{-x}(e^x - 1) = 5(1 - e^{-x})$$

Solving by iteration for x gives $x \approx 4.965$. Returning to the definition of x ,

$$4.965 = \frac{hc}{k_B T \lambda_{max}} \rightarrow \lambda_{max} \approx \frac{2.989 \cdot 10^{-3}}{T}$$

This is Wien's law, commonly written $\lambda = b/T$, where $b \approx 2.989 \cdot 10^{-3}$.

- [3] **Problem 5.** The luminosity of a secondary body depends on the visible illuminated area of the object. In this question we briefly study that phenomenon.

a) Consider the following situation:

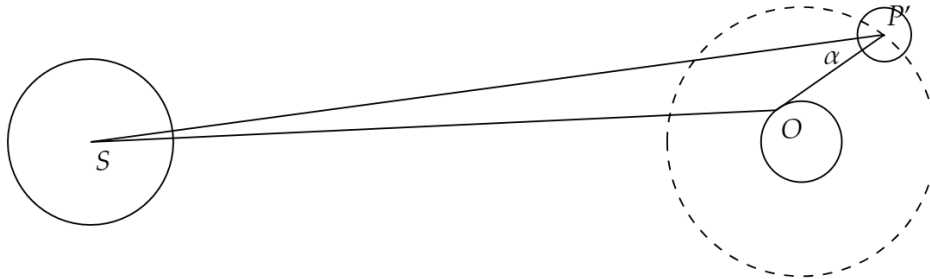


Figure 1: Sun-Earth-Moon diagram

Find an expression for the ratio Φ of the illuminated area, as a function of α , to the total area of the planet.

- b) Find the angles α at which the Moon is in the phases new, waxing, full, and waning, respectively.

Solution

To solve the problem, follow the diagram below:

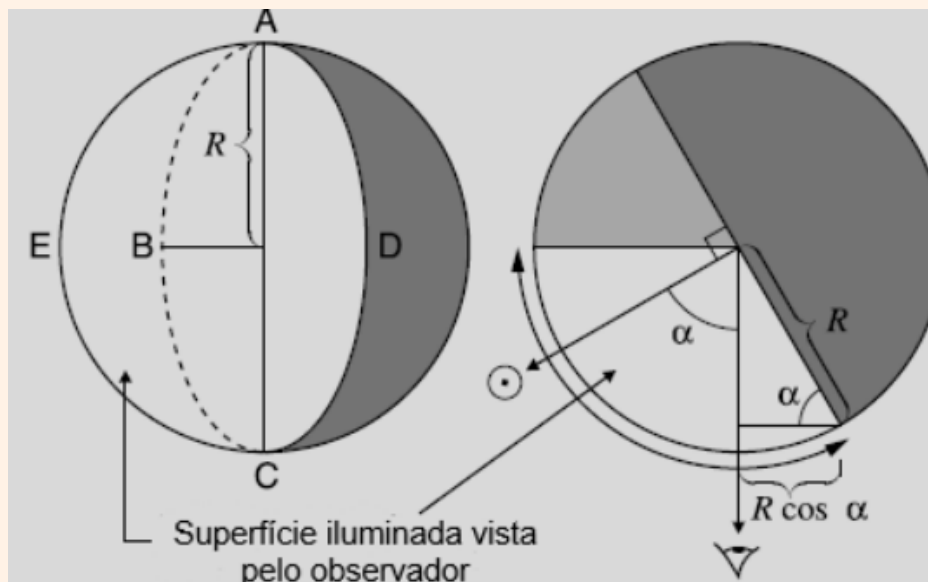


Figure 2: Source: Introduction to Planetary Photometry

On the left is a sketch of how the planet appears in the sky, and on the right a view "from

above". The area we see is half the area of the circle plus half the area of an ellipse with semi-major axis R and semi-minor axis $R \cos \alpha$. Thus

$$\Phi = \frac{\pi R^2/2 + \pi R^2 \cos \alpha/2}{\pi R^2} \quad \therefore \Phi = \frac{1 + \cos \alpha}{2}$$

At new moon, $\Phi = 0$; at waxing, $\Phi = 1/2$; at full moon, $\Phi = 1$; and at waning, $\Phi = 1/2$. Thus

$$\alpha_{nova} = 180^\circ, \alpha_{crescente} = 90^\circ, \alpha_{cheia} = 0^\circ, \alpha_{minguante} = 90^\circ$$

- [5] **Problem 6.** (Magna notes) In this problem we model the effect of the atmosphere on Earth. Suppose that the Sun is a blackbody of temperature T_1 and radius R_1 . Earth is a sphere located at a distance R from the Sun, with radius R_3 . Earth's emissivity is ϵ_3 .

- a) If Earth had no atmosphere, determine its equilibrium temperature, T_3 .
- b) Now we include the atmosphere. Model it as a spherical shell of gas, with emissivity ϵ_2 and outer radius $R_2 > R_3$, concentric with Earth. In thermal equilibrium, its absorptivity at ultraviolet and infrared wavelengths is ϵ_2 . The atmosphere transmits a fraction t of the ultraviolet radiation but is completely opaque to infrared radiation. Assuming that the Sun emits ultraviolet light while Earth emits and re-emits in the infrared, determine the temperature T_2 of the atmosphere and the temperature T_3 of Earth in thermodynamic equilibrium. Assume that the atmosphere is a perfect thermal conductor, so that all radiation incident on it is distributed uniformly over its surface.

Solution

- a) In thermodynamic equilibrium, the power absorbed equals the power emitted. The emissivity and the absorptivity are also equal. Thus

$$\frac{4\pi R_1^2 \sigma T_1^4}{4\pi R^2} \pi R_3^2 \epsilon_3 = 4\pi R_3^2 \sigma T_3^4 \epsilon_3$$

Solving for T_3 ,

$$T_3 = T_1 \left(\frac{R_1^2}{4\pi R^2} \right)^{1/4}$$

- b) The temperature of the atmosphere is

$$T_2 = T_1 \left(\frac{R_1^2}{4\pi R^2} \right)^{1/4}$$

The reasoning is analogous to the previous part. The luminosity transmitted by Earth is

$$L_2 = 4\pi R_2^2 \sigma T_2^4 \epsilon_2 t$$

Earth reflects all of this radiation, and the atmosphere reflects only a fraction t back, so the total luminosity is

$$L_{2,T} = 4\pi R_2^2 \sigma T_2^4 \epsilon_2 (t + t^2 + t^3 + \dots)$$

The expression in parentheses is an infinite geometric series, with first term $q_1 = t$ and common ratio $r = t$. Since $t < 1$, the sum is

$$(t + t^2 + t^3 + \dots) = \sum_{n=1}^{\infty} t^n = \frac{t}{1-t}$$

Equating the two luminosities,

$$L_{2,T} = L_3 \rightarrow \frac{4\pi R_2^2 \sigma T_2^4 \epsilon_2 t}{1-t} = 4\pi R_3^2 \sigma T_3^4$$

Solving for T_3 ,

$$T_3 = T_2 \left(\frac{R_2^2 \epsilon_2}{R_3^2} \frac{t}{1-t} \right)^{1/4}$$

Substituting T_2 ,

$$T_3 = T_1 \left(\frac{R_1^2 R_2^2}{4\pi R^2 R_3^2} \frac{\epsilon_2 t}{1-t} \right)^{1/4}$$

- [2] **Problem 7.** (Problem set 2 - 2021) The Ring Nebula (M57) has an apparent magnitude of 9 and an angular diameter of $2'$ for an observer on Earth. What would be the apparent magnitude of the night sky of a planet orbiting a star exactly at the center of M57?

Solution

The night sky covers a solid angle of 2π sr, while the solid angle we see is $\Omega \approx \pi\theta^2$, where θ is the angular radius. Since the flux is proportional to Ω ,

$$m_{ceu} - m = -2.5 \log \left(\frac{\Omega_{ceu}}{\Omega} \right)$$

To convert Ω from arcmin² to sr, multiply by $(\frac{\pi}{180 \cdot 60})^2$. Thus

$$m_{ceu} = 9 - \log \left(\frac{2\pi}{\pi(\frac{\pi}{180 \cdot 60})^2} \right)$$

$$m_{ceu} = -9.43$$

- [4] **Problem 8.** (Adapted from Problem set 4 - 2021) The Cherenkov effect was first detected by the Soviet scientist Pavel Cherenkov in 1937. Later, together with his colleagues I. E. Tamm and I. M. Frank, he gave the phenomenon a physical interpretation and was awarded the Nobel Prize in Physics in 1958. Before a mathematical treatment, we first need to understand the principle a little better.

When high-energy charged particles travel through a dielectric medium, atoms can be excited if the particle speed exceeds the phase velocity ($\frac{c}{n}$). Those atoms, in turn, emit electromagnetic radiation when they return to the ground state. The emitted waves spread spherically and, when superposed, form a cone of opening angle 2α , as the figure below shows.

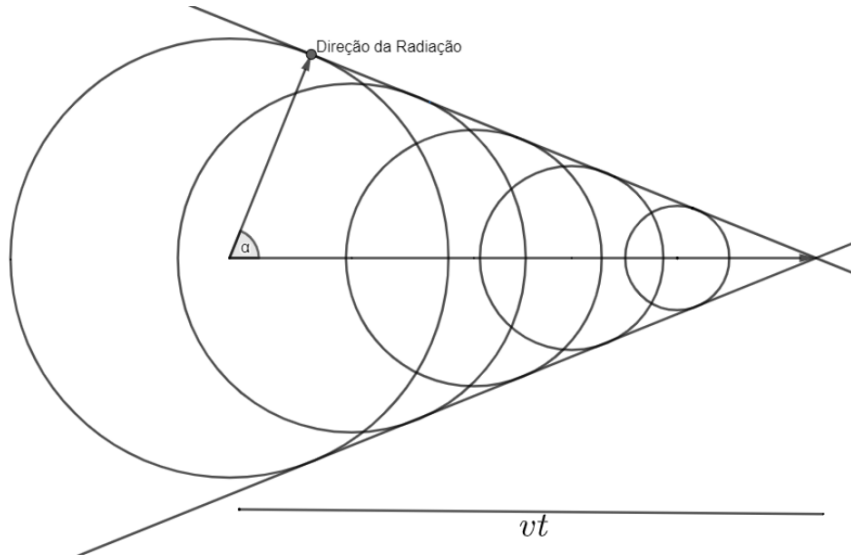


Figure 3: Radiation mechanism of the Cherenkov effect

This effect is similar to a jet moving at supersonic speed: it follows the same principle as the Mach cone, but with light. We now develop the mathematical model of the Cherenkov effect.

Part A - Theoretical Model

Consider a particle moving at relativistic speeds through a medium of refractive index n . Its rest mass is m_0 , and it has linear momentum p and speed v . At a given moment, a photon is emitted at an angle α , as Figure 1 shows.

- Letting μ be the frequency of the emitted photon, determine its linear momentum, p_μ , and its energy, E_μ . Your answer must be in terms of n , μ , and physical constants.
- Find an expression for the linear momentum of the particle after it emits the photon, in terms of p_μ , p , and α .
- Letting $\beta_n = \frac{c}{vn}$, prove that the relation below holds:

$$\cos \alpha = \frac{1}{\beta_n} \quad (1)$$

- d) Assuming that linear momentum and energy are conserved, determine the minimum speed at which the Cherenkov effect can occur. **Hint:** Compared with the other parameters, the factor $(n^2 - 1)h\mu$ may be neglected.

Part B - Nuclear Reactions

The proton-proton chain is a sequence of fusion reactions that convert hydrogen into helium. One possible branch of the proton-proton chain is *pp* IV, in which, in principle, a helium-3 atom reacts directly with a proton, according to the reaction below:



- e) State the nuclear conservation law you use, and identify the missing particle in the blank of the reaction above.
- f) State the nuclear conservation law you use, and identify the missing particle in the blank of the reaction below:

$$\pi^- \rightarrow \mu^- + \text{---} \quad (3)$$

Data: Pion mass: $140 \text{ MeV}/c^2$, muon mass: $106 \text{ MeV}/c^2$.

Solution

Part A - Theoretical Model

- a) For a photon, the de Broglie relations give

$$E = h\mu, \quad p = \frac{h}{\lambda}$$

where h is Planck's constant.

Using the fundamental wave relation, $\lambda = v/\mu = \frac{c}{n\mu}$, so

$$E = h\mu, \quad p = \frac{nh\mu}{c}$$

- b) Since the total momentum is conserved, suppose the particle moves with momentum p along the x axis before emitting a photon. Then

$$p = p_\mu \cos \alpha + p'_x$$

$$p_\mu \sin \alpha = p'_y$$

$$p' = \sqrt{p_x^2 + p_y^2}$$

Solving for p' ,

$$p'^2 = (p - p_\mu \cos \alpha)^2 + p_\mu^2 \sin^2 \alpha$$

$$p'^2 = p^2 - 2p_\mu p \cos \alpha + p_\mu^2$$

Finally,

$$p' = \sqrt{p^2 - 2pp_\mu \cos \alpha + p_\mu^2}$$

c) Consider the following figure.

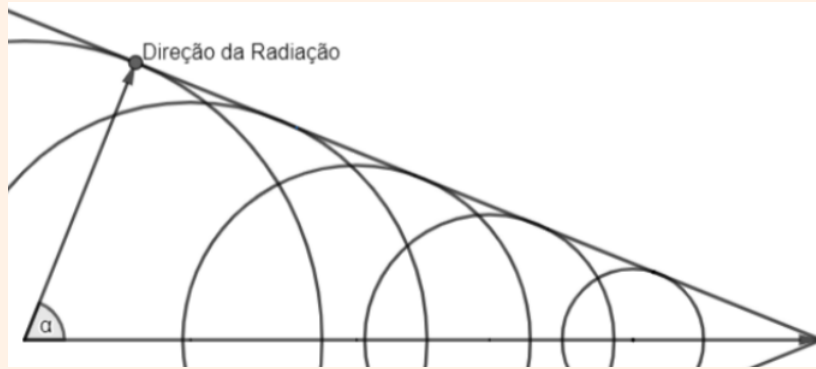


Figure 4: Cherenkov diagram

The radius of the circle is the distance traveled by the photon. The distance from the center of the circle to the end of the cone is vt , so

$$\cos \alpha = \frac{ct}{nvt} = \frac{c}{nv} \equiv \frac{1}{\beta_n}$$

d) Setting $c = 1$, energy conservation gives

$$\sqrt{p^2 + m_0^2} = \sqrt{p'^2 + m_0^2} + h\mu$$

Working through this expression,

$$\sqrt{p'^2 + m_0^2} = \sqrt{p^2 + m_0^2} - h\mu$$

$$p'^2 + m_0^2 = p^2 + m_0^2 + h^2\mu^2 - 2h\mu\sqrt{p^2 + m_0^2}$$

Substituting p' ,

$$p^2 - 2pp_\mu \cos \alpha + p_\mu^2 = p^2 + h^2\mu^2 - 2h\mu\sqrt{p^2 + m_0^2}$$

Solving for $\cos \alpha$,

$$\cos \alpha = \frac{p_\mu^2 - h^2\mu^2 + 2h\mu\sqrt{p^2 + m_0^2}}{2pp_\mu}$$

Simplifying,

$$\cos \alpha = \frac{p_\mu}{2p} - \frac{h^2\mu^2}{2pp_\mu} + \frac{h\mu\sqrt{p^2 + m_0^2}}{pp_\mu}$$

Substituting $p_\mu = nh\mu$,

$$\cos \alpha = \frac{nh\mu}{2p} + \frac{h\mu}{2np} + \frac{\sqrt{p^2 + m_0^2}}{np}$$

$$\cos \alpha = \frac{h\mu(n^2 - 1) + 2\sqrt{p^2 + m_0^2}}{2np}$$

Neglecting the term $h\mu(n^2 - 1)$ and substituting $\cos \alpha$,

$$\frac{1}{nv} = \frac{\sqrt{p^2 + m_0^2}}{np}$$

Solving for v ,

$$v = \frac{p}{\sqrt{p^2 + m_0^2}}$$

Finally, restoring the factors of c ,

$$v = \frac{pc^2}{\sqrt{p^2c^2 + m_0^2c^4}}$$

Part B - Nuclear Reactions

- e) Using charge conservation, the left-hand side has $2 + 1 = 3$ protons, but the right-hand side has only 2, so the missing particle must contain one proton. The mass number is $3 + 1 = 4$ on the left and 4 on the right. The missing particle must therefore have the charge of a proton and a mass much smaller than a proton's. The only particle that fits this description is the positron, e^+ .

- f) Lepton number is conserved. The pion is not a lepton, $n_L = 0$, but the muon is a lepton with $n_L = +1$, so the missing particle must have $n_L = -1$. The only neutral antilepton associated with the muon and carrying $n_L = -1$ is the muon antineutrino, $\bar{\nu}_\mu$. The complete reaction is therefore

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

[3] **Problem 9.** (Problem set 3 - Vinhedo 2022) Juvelino, observing directly from his observatory in Paris, France, monitors the star Polaris (α UMi). His goal is to find the color temperature T_c of the star. Some of the data he has on his target are:

- Apparent magnitude in the V band: $V = 1.98$;
- Absolute magnitude in the V band: $M_V = -3.60$;
- Absolute magnitude in the B band: $M_B = -3.19$.

Using the information provided, help Juvelino!

- a) From many observations, Juvelino determined that the interstellar extinction in the V band toward Polaris is $a_V = 5.8 \text{ mag/kpc}$. Find the distance, in pc, from α UMi to Earth.

- b) Using the empirical relation

$$\frac{A_V}{E_{B-V}} = 3.0 \quad (4)$$

where A_V is the total interstellar extinction in the V band and E_{B-V} is the $B - V$ color excess, find the $B - V$ color index of the observed star.

- c) Show the relation

$$T_c = \frac{7009}{(B - V) + 0.47} \quad (5)$$

in which the color temperature is given in kelvin. To do so, use the fact that stars of spectral class $A0$ have $(B - V) = 0$ and $T_c = 15000 \text{ K}$. Also use the wavelengths of the B and V bands, $\lambda_B = 440 \text{ nm}$ and $\lambda_V = 548 \text{ nm}$, respectively. Justify any approximations you make.

HINT: Planck's law may be useful

- d) Find the color temperature of Polaris.

Solution

- a) The expression that correctly relates the magnitudes is

$$V - M_V = 5 \log d - 5 + a_V d$$

This equation can be solved only by iteration:

$$d = \frac{V - M_V - 5 \log d + 5}{a_v}$$

Iterating, we obtain $d \approx 100 \text{ pc}$.

b) By definition, $E_{B-V} = A_B - A_V$. Thus

$$\frac{A_V}{A_B - A_V} = 3.00$$

$$A_B = \frac{A_V}{3} + A_V = 0.77$$

Thus

$$B - M_B = 5 \log d - 5 + A_B \rightarrow B = 2.58$$

By definition, the $B - V$ index is $U_{B-V} = B - V = 0.60$

c) Using Planck's law,

$$\frac{B_B}{B_V} = \frac{F_B}{F_V} \left(\frac{\lambda_V}{\lambda_B} \right)^5 \frac{e^{\frac{\beta hc}{\lambda_V} - 1}}{e^{\frac{\beta hc}{\lambda_B} - 1}}$$

Using the Wien approximation,

$$\frac{F_B}{F_V} = \left(\frac{\lambda_V}{\lambda_B} \right)^5 \frac{e^{\frac{\beta hc}{\lambda_V}}}{e^{\frac{\beta hc}{\lambda_B}}}$$

Using Pogson's equation to compare the magnitudes,

$$B - V = -2.5 \log \left(\frac{F_B}{F_V} \right) + C$$

A constant is included because the two bands have different wavelengths. Working through the expression,

$$B - V = -2.5 \log \left(\left(\frac{\lambda_V}{\lambda_B} \right)^5 \frac{e^{\frac{\beta hc}{\lambda_V}}}{e^{\frac{\beta hc}{\lambda_B}}} \right) + C$$

Working through it further,

$$B - V = -2.5 \log \left(\frac{\lambda_V}{\lambda_B} \right)^5 + 2.5 \beta hc \left(\frac{1}{\lambda_B} - \frac{1}{\lambda_V} \right) \log e + C$$

Using $\beta = \frac{1}{k_B T_c}$ and substituting the values,

$$B - V = -1.19 + \frac{7009}{T_c} + C$$

For type A_0 , $(B_V) = 0$ and $T_c = 15000\text{K}$. Thus

$$-1.19 + \frac{7009}{15000} + C = 0 \rightarrow C = 0.72$$

Thus

$$B - V = -0.47 + \frac{7009}{T_c} \rightarrow T_c = \frac{7009}{(B - V) + 0.47}$$

d) From the previous equation,

$$T_c = \frac{7009}{0.60 + 0.47} \approx 6600 \text{ K}$$

[4] **Problem 10.** (Problem set 4 - 2020) Frightening as it is, Juvelino's banana plantation has a perfect sky for one of his two passions: astrophotography. He has a telescope 200 mm in diameter with a super CCD attached. Its parameters are: pixel side $5 \mu\text{m}$, plate scale 3.2 arcmin/mm, quantum efficiency 97%, RON (read-out noise) 1 count/pixel, and DN (thermal noise) 1 count/(pixel.hour). He prefers to observe in the V band (central wavelength $\lambda_V = 5500 \text{ \AA}$, bandwidth $\Delta\lambda = 820 \text{ \AA}$), because he remembers a magic number associated with it: the photon flux of a magnitude-0 object in this band is $\phi_0 = 1000 \text{ photons}/(\cdot \text{ cm}^2)$. Juvelino has two favorite targets: a star of magnitude $m_V = 4.51$ and a globular cluster of uniform surface brightness in V of $19.5 \text{ mag/arcsec}^2$ and angular diameter $\theta = 40''$.

a) The night-sky brightness at Juvelino's farm is $21.9 \text{ mag/arcsec}^2$ in the V band, and the astronomical seeing is $\theta_0 = 1.2''$. Calculate the signal-to-noise ratio obtained in a photo of his favorite star with exposure time $t_{exp} = 30\text{s}$.

b) Calculate the zero-point magnitude of Juvelino's telescope + CCD system, using the data for the star.

Juvelino's other great passion is Jiang, a Chinese university student who lives in Beijing. After years apart, Jiang suggests that Juvelino move to China. Before deciding, he looks up the night-sky brightness in the V band in his beloved's city, which is $19.3 \text{ mag/arcsec}^2$. That will affect imaging of the cluster, which is also visible from Jiang's home.

c) Juvelino cannot live without photographing the cluster. He has inviolable rules for imaging this object, wherever he does it: reach a signal-to-noise ratio $S/N = 10$, and take the same number X of photos of the cluster in one night. X is the number of photos allowed by the sky conditions at his banana plantation. That number of frames per night is limited by the CCD battery (which lasts 3.2 hours) and by the exposure time required for each photo. Help him with this dilemma! Do the calculations, find how many photos of the cluster he can take in one night (in Beijing and at the farm), and conclude: will he see his beloved again, or will this love be too noisy?

Solution

The official solution can be found [here](#).